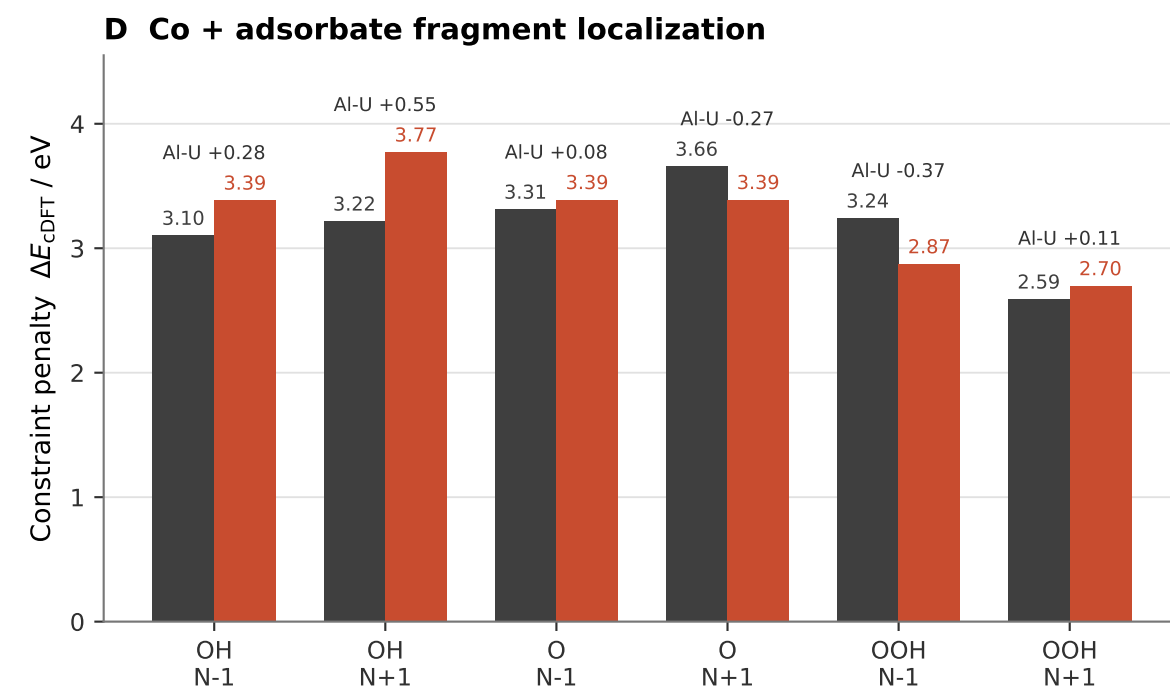
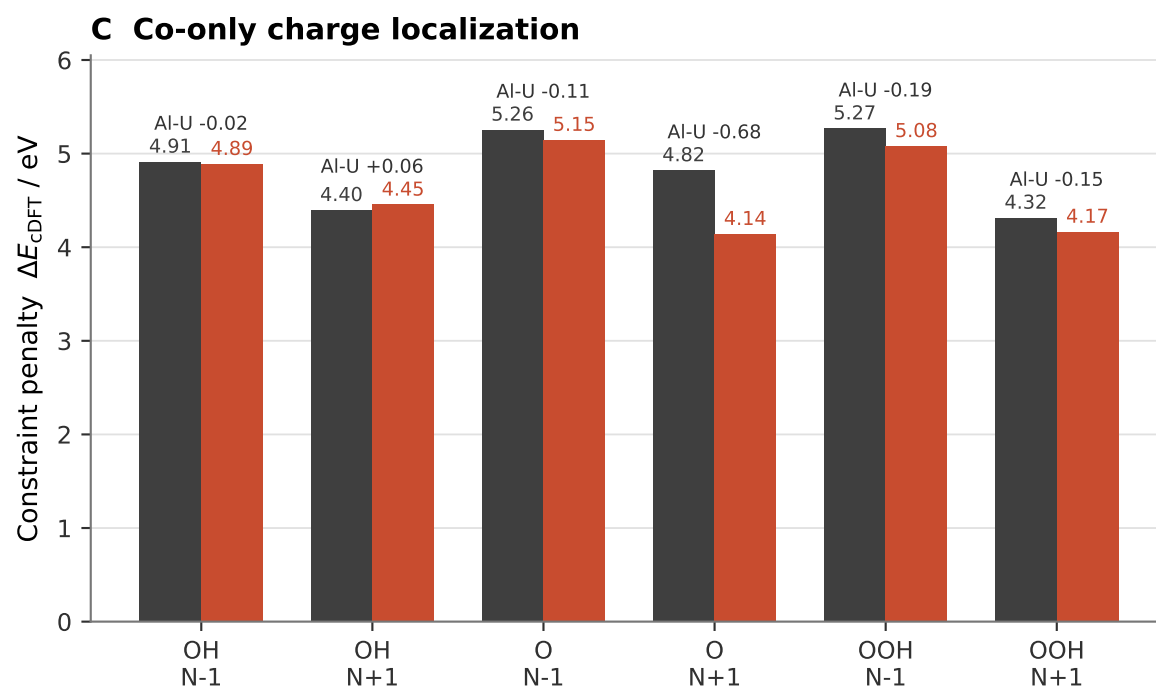
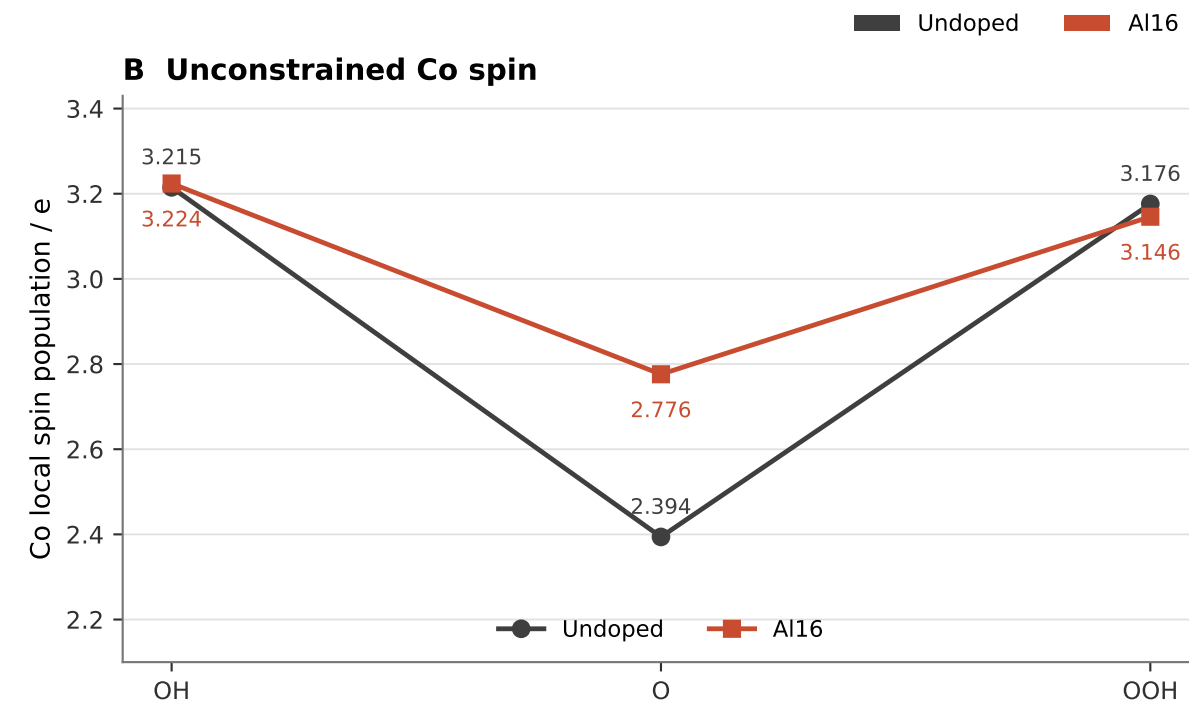
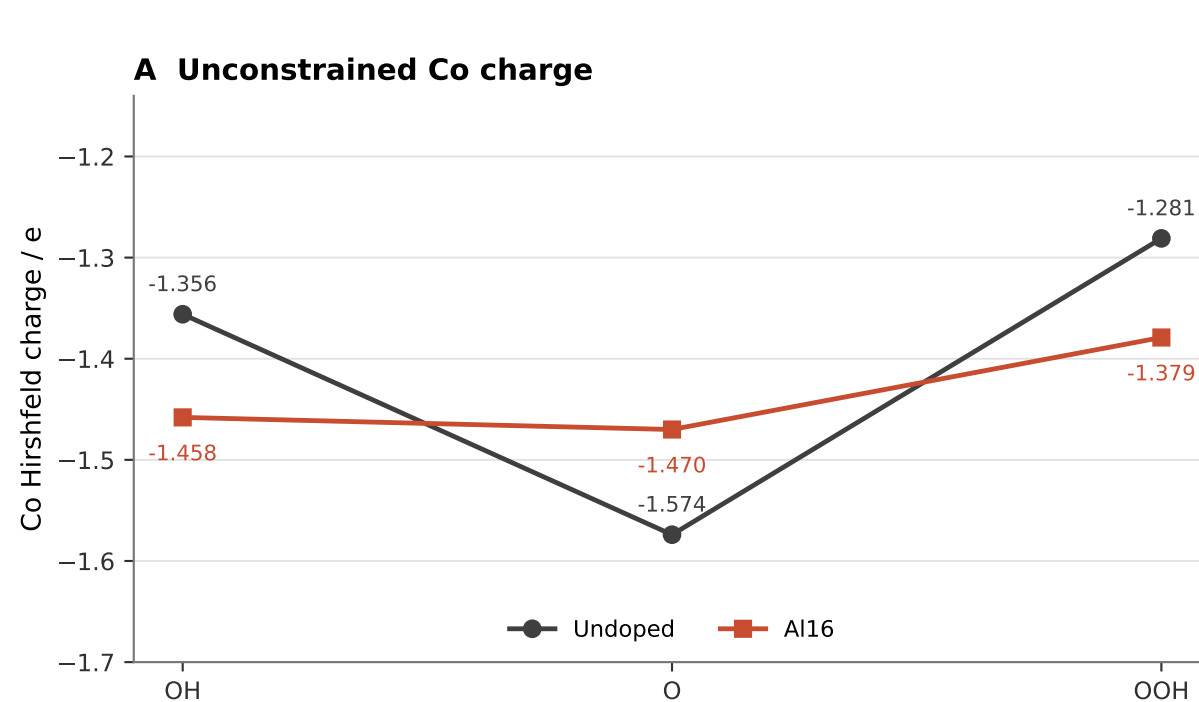


Complete fixed-geometry cDFT comparison: undoped vs Al16



N-1 removes one electron (hole localization); N+1 adds one electron. Values are localization penalties relative to each unconstrained state, not OER step free energies. Models are historical accepted undoped Co32 and chemically valid Al16 Co40; local sites are not identical, so use differences as mechanistic trends rather than formal oxidation states.